1 Filter

- Low pass filtering: $s = \frac{a+b}{2}$ (average)
- High pass filtering: $d = a - s$ (difference)

2 Problem A

2.1 Part (i)

Original data sequence: $[71, 67, 24, 26, 36, 32, 14, 18]$

2.1.1 Level 1 Transformation

- $s_1 = [(\frac{71+67}{2}), (\frac{24+26}{2}), (\frac{36+32}{2}), (\frac{14+18}{2}]) = [69, 25, 34, 16]$ (cA)
- $d_1 = [(71 - 69), (24 - 25), (36 - 34), (14 - 16)] = [2, -1, 2, -2]$ (cD)

$\Rightarrow v_1 = [s_1, d_1] = [69, 25, 34, 16, 2, -1, 2, -2]$

2.1.2 Level 2 Transformation

- $s_2 = [(\frac{69+25}{2}), (\frac{34+16}{2})] = [47, 25]$
- $d_2 = [(69 - 47), (34 - 25)] = [22, 9]$

$\Rightarrow v_2 = [s_2, d_2] = [47, 25, 22, 9]$

2.1.3 Level 3 Transformation

- $s_3 = [(\frac{47+25}{2})] = [36]$
- $d_3 = [(47 - 36)] = [11]$

$\Rightarrow v_3 = [s_3, d_3] = [36, 11]$

2.1.4 Wavelet Transform Result

Level	Low Pass Filtering (s)	High Pass Filtering (d)
1	$[69, 25, 34, 16]$	$[2, -1, 2, -2]$
2	$[47, 25]$	$[22, 9]$
3	$[36]$	$[11]$

2.2 Part (ii)

2.2.1 Inverse Coefficients Formula

From $s = \frac{a+b}{2}$ and $d = a - s$:

- $a = s + d$
- $b = s - d$

2.2.2 Inverse Wavelet Transform

Input data: $[s_3, d_3, d_2, d_1] = [36, 11, 22, 9, 2, -1, 2, -2]$

$$
\begin{aligned}
[[36+11], [36-11], 22, 9, 2, -1, 2, -2] &= [47, 25, 22, 9, 2, -1, 2, -2] \\
[[47, 25] + [22, 9], [47, 25] - [22, 9], 2, -1, 2, -2] &= [[69, 34], [25, 16], 2, -1, 2, -2] \\
&= [69, 25, 34, 16, 2, -1, 2, -2] \\
[[69, 25, 34, 16] + [2, -1, 2, -2], [69, 25, 34, 16] - [2, -1, 2, -2]] &= [[71, 24, 36, 14], [67, 26, 32, 18]] \\
&= [71, 67, 24, 26, 36, 32, 14, 18]
\end{aligned}
$$

3 Problem B

3.1 Part (i)

Original data sequence: $[18, 14, 32, 36, 26, 24, 67, 71]$

3.1.1 Level 1 Transformation

- $s_1 = [(\frac{18+14}{2}), (\frac{32+36}{2}), (\frac{26+24}{2}), (\frac{67+71}{2}]) = [16, 34, 25, 69]$ (cA)
- $d_1 = [(18-16), (32-34), (26-25), (67-69)] = [2, -2, 1, -2]$ (cD)

$\Rightarrow v_1 = [s_1, d_1] = [16, 34, 25, 69, 2, -2, 1, -2]$

3.1.2 Level 2 Transformation

- $s_2 = [(\frac{16+34}{2}), (\frac{25+69}{2})] = [25, 47]$
- $d_2 = [(16-25), (25-47)] = [-9, -22]$

$\Rightarrow v_2 = [s_2, d_2] = [25, 47, -9, -22]$

3.1.3 Level 3 Transformation

- $s_3 = [(\frac{25+47}{2})] = [36]$
- $d_3 = [(25 - 36)] = [-11]$

$\Rightarrow v_3 = [s_3, d_3] = [36, -11]$

3.1.4 Wavelet Transform Result

Level	Low Pass Filtering (s)	High Pass Filtering (d)
1	$[16, 34, 25, 69]$	$[2, -2, 1, -2]$
2	$[25, 47]$	$[-9, -22]$
3	$[36]$	$[-11]$

3.2 Part (ii)

3.2.1 Inverse Coefficients Formula

From $s = \frac{a+b}{2}$ and $d = a - s$:

- $a = s + d$
- $b = s - d$

3.2.2 Inverse Wavelet Transform

Input data: $[s_3, d_3, d_2, d_1] = [36, -11, -9, -22, 2, -2, 1, -2]$

$$
\begin{aligned}
[[36 + (-11)], [36 - (-11)], -9, -22, 2, -2, 1, -2] &= [25, 47, -9, -22, 2, -2, 1, -2] \\
[[25, 47] + [-9, -22], [25, 47] - [-9, -22], 2, -2, 1, -2] &= [[16, 25], [34, 69], 2, -2, 1, -2] \\
&= [16, 34, 25, 69, 2, -2, 1, -2] \\
[[16, 34, 25, 69] + [2, -2, 1, -2], [16, 34, 25, 69] - [2, -2, 1, -2]] &= [[18, 32, 26, 67], [14, 36, 24, 71]] \\
&= [18, 14, 32, 36, 26, 24, 67, 71]
\end{aligned}
$$